strongly connected? An important theorem by the famous mathematician Robbins proved that every 2-edge connected graph can have a strongly connected directed graph. A graph is said to be 2-edge connected if it remains connected even if any one edge is removed. If the given graph is not 2-edge connected, the orientation returned by the *strong_orientation( )* method will ensure that each *2-edge connected* component has a strongly connected orientation. Please verify that there is a directed path between every pair of vertices in *Rgraph_13*, shown in Figure 4, before you proceed further.

In the last two articles in this series, we dealt with undirected multigraphs. Similarly, there are directed multigraphs in which multiple directed edges can exist between a pair of vertices. Let us now see an example of directed multigraphs. The program '*dmultigraph14.sagews*' shown below generates a directed multigraph called *Dmultigraph_14*.

```
dirmg = DiGraph(multiedges=True, loops=True)
dirmg.add_vertices([0, 1, 2, 3])
edges = [(0, 1), (0, 1), (1, 2), (1,3), (2, 3), (2, 2), (3, 
0), (3, 1)]
dirmg.add_edges(edges)
dirmg.show(edge_labels=False)
```

Upon executing the program '*dmultigraph14.sagews*' using CoCalc, we obtain a directed multigraph called *Dmultigraph_14*, which is shown in Figure 5. From Figure 5, it can be observed that there are two edges between vertices 1 and 3 (one from vertex 1 to vertex 3 and the other from vertex 3 to vertex 1), as well as two edges between vertices 0 and 1 (both directed from vertex 0 to vertex 1).

Now, let us revisit the examples we have discussed to understand the use of undirected and directed graphs in modelling real-world scenarios. We used undirected graphs to represent a city street map where all the roads allow two-way transportation and directed graphs to represent a city street map where all the roads allow only one-way traffic. However, in reality, no city has such a uniform road system. Every city has some roads that allow two-way traffic and others that only allow one-way traffic. How can we capture the features of such a city street map and use graph-theoretic methods to elicit information? Mixed graphs and mixed multigraphs are handy in this scenario. *Mixed graphs* and *mixed multigraphs* can have both directed and undirected edges. In the example of the city street map, the undirected edges will represent roads that allow two-way traffic, and the directed edges will represent roads that allow only one-way traffic.

However, we face a slight hiccup when using SageMath to simulate mixed graphs and mixed multigraphs. The problem is that SageMath does not have a built-in class to handle mixed graphs and mixed multigraphs directly. Instead, we can use directed multigraphs to simulate mixed graphs and mixed multigraphs, which allow parallel directed edges. This is helpful because an undirected edge between vertices, say *A* and *B*, is equivalent (for almost all practical purposes) to two directed edges: one from vertex *A* to vertex *B* and another from vertex *B* to vertex *A*. This double directed edge construction is a classical technique used in graph theory to construct mixed graphs and mixed multigraphs using directed multigraphs.

Let us see a practical example of representing a mixed graph with a directed multigraph. How can we represent the mixed graph with four vertices shown in Figure 6 using a directed multigraph?

The program '*dmultigraph15.sagews*' shown below generates a directed multigraph called *Dmultigraph_15*, which simulates the mixed graph shown in Figure 6.

```
DG = DiGraph( )
DG.add_vertices([0, 1, 2, 3])
edges = [(0, 1), (1, 2), (2, 1), (2, 3), (3, 0), (0, 3)]
DG.add_edges(edges)
DG.show( )
```

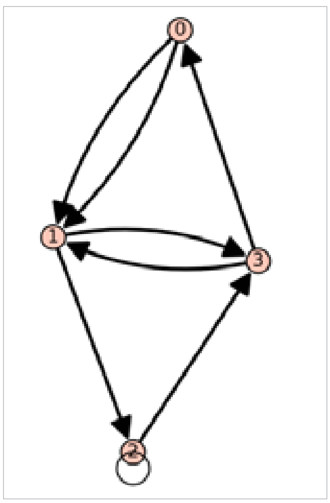

Figure 5: The directed multigraph *Dmultigraph_14*

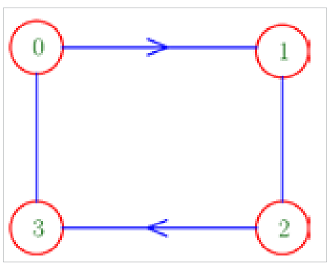

Figure 6: A mixed graph on four vertices

Upon executing the program '*dmultigraph15.sagews*' using CoCalc, we obtain a directed multigraph called *Dmultigraph_15*, which is shown in Figure 7. Observe Figure 7 carefully and verify that *Dmultigraph_15* indeed simulates the mixed graph shown in Figure 6. Also, keep in mind that to simulate mixed multigraphs, we can use directed multigraphs with multiple directed edges in the same direction, like the two directed edges from vertex 0 to vertex 1 in *Dmultigraph_14* shown in Figure 5. Try to draw a mixed multigraph and then simulate it using a directed multigraph as an exercise.

Now, it is time to wrap up our discussion. First, let me answer the question